

Roll No.

TCE-813

B. TECH. (CIVIL ENGINEERING) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

HYDROPOWER ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the strength and opportunity of a water-power in India. (CO1)
- (b) A power development site has the average estimated river flows as given in the table below (flow is million m³) : (CO1)

Months	Flow
January	68
February	35
March	77
April	53

P. T. O.

May	15
June	17
July	35
August	89
September	92
October	44
November	54
December	87

Calculate the required capacity of the reservoir in order to produce 14500 kW continuously throughout the year at an overall efficiency of 85% under a head of 110 m.

- (c) Predict the load of a town in year 2021 using Scheer's formula and the following data : (CO1)

Per capita load in year 2018 = 450 kWh

Population growth rate = 8% (constant for five years period)

Population in year 2018 = 60,00,000

2. (a) Run-of-river plant on a stream has inflow of 20 cumecs and net head of 30 m with provision for pondage to meet daily peak demand with a load factor of 60%. Calculate : (CO2)

(i) The power generation capacity of the plant at 80% overall efficiency.

(ii) The plant runs as peaking station for 3 hrs. and balance period in the day for average load.

What amount of pondage is needed ?

- (b) The turbine is to operate under a head of 24 m at 200 r. p. m. the discharge is $8.5 \text{ m}^3/\text{s}$. If the overall efficiency is 88%, calculate : (CO2)
- (i) Power generated
 - (ii) Specific speed of turbine
 - (iii) Type of turbine
- (c) Explain run-of-river plants. Also describe the general arrangements of the run-of-river plants. (CO2)
3. (a) Describe all the three main divisions of a power station with a neat sketch. (CO3)
- (b) "Some of the power plants adopt two other types of super-structure in which the machine hall concept does not exist." Explain the types of super-structure. (CO3)
- (c) Describe all the ways of locating an underground power station. Explain with neat sketch. (CO3)
4. (a) Determine the expressions for the water hammer pressure in case of a rigid pipe and in case of an elastic pipe. (CO4)
- (b) A hydro electric station is not to be supplied with $10 \text{ m}^3/\text{sec}$ of water through a penstock which has friction factor of 0.016. The maximum normal head of the penstock is 50 kg/cm^2 and a water hammer over pressure of 20% over the normal pressure is anticipated. The safe head in the steel used is presumed to be 3000 kg/cm^2 . The ready penstocks at site are likely to cost ₹ 15,000 per tonne including erection charges. The life of the project is 50 years and the rate of interest is 7 percent. It is proposed to sell the energy at the rate of Re. 0.03/kWh. Predict the optimum diameter of the penstock, given the OMR costs are 5 percent. Assume the turbine efficiency to 90 percent and the annual load factor is 0.4. (CO4)

- (c) Describe the 'economical diameter of a penstock'. Also explain the different methods of calculating the economical diameter of penstock.

(CO4)

5. (a) Describe the main types of turbine. Draw the diagram of the runners of various turbines. (CO5)

- (b) At the Volga Imeni V. I. Lenin hydroelectric plant, in USSR, the Kaplan turbine used has the following data :

Operation head is 22.5 m, Output power at this head is 126 MW, Discharge at this head is $616 \text{ m}^3/\text{sec}$, Speed is 68.2 r.p.m., Runner tip-to-tip diameter (D) is 9.3 m, Hub diameter (D_h) is 4.3 m, Number of blades is 6.

Calculate the speed ratio, the flow ratio and the overall efficiency and the maximum suction draft head. (CO5)

- (c) A Francis turbine on a major hydro-power project in Pakistan is to operate under a net head of 150 m. The desired output per turbine is 175 MW and the synchronous generator speed is 136.4 r.p.m. Model studies indicate an overall efficiency of 91.5 percent and the prototype efficiency is likely to be at least 0.5 percent. Hyd. efficiency is estimated as 96 percent. Calculate the specific speed and the discharge to the machine. Also suggest suitable values of exist angle of guide vanes, the vane angles of the wheel, the inlet and outlet diameter of the wheel and the breadth of the wheel at the inlet. Make suitable assumptions. (CO5)

Roll No.

TCE-810

**B. TECH. (CE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022**

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How would you differentiate sociological and technological disaster with examples ? (CO1)
- (b) How would you describe 'Hazards' and 'Vulnerability' ? Write *five* possible causes of forest fires ? (CO1)
- (c) How would you contrast the impact of disaster on victims and on the society (not on the nature) ? (CO1)

P. T. O.

2. (a) Can you make use of the fact that the man-made disasters have a long-term impact than a natural disaster on the environment ? Explain.(CO2)
- (b) How would you contrast the impact of climate change on disaster in terms of declining biodiversity, human migration and decrease in agricultural yield ? (CO2)
- (c) What elements would you choose to explain differential impacts of a disaster ? (CO2)
3. (a) What facts would you suggest explaining positive realms of a disaster on the development. (CO3)
- (b) What elements can you list to explain prediction, forecasting and warning in context to disaster management ? (CO3)
- (c) Explain DPP and the elements and activities associated with DPP. (CO3)
4. (a) Are you aware with the functions of the rescue party and importance of various types of rescue training. (CO4)
- (b) Explain Immediate relief assessment survey. Various stages of assessment and what kind of data is collected under assessment survey ? (CO4)
- (c) Under post disaster management kindly highlight the urban risks and urban disaster mitigation strategies in detail. (CO4)
5. (a) Necessity of DMA. List down the government bodies that are associated with DMA. (CO5)

(3)

(b) Explain the following :

(CO5)

(i) Environment protection act

(ii) NEC

(iii) NDRF

(iv) DDMA

(c) Consider a case study of Flash floods in Uttarakhand and being the head of NEC what will be the flowchart that you will follow to mitigate this disaster.

(CO5)

Roll No.

TCE-801

**B. TECH. (CE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
EARTHQUAKE RESISTANT DESIGN OF BUILDINGS**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the equation of motion for a single degree of freedom free undamped system. (CO1)

(b) An undamped SDOF system is subjected to harmonic excitation. Resonance occurs when exciting frequency is 6 rad/s. When the mass is increased by 1 kg resonance is found to occur at 5.86 rad/s. Find the original mass and stiffness of the system. (CO1)

OR

(c) Determine natural frequency of mass (m) placed at one end of cantilever beam of length (l) and flexural rigidity (EI) having negligible mass.

(CO1)

P. T. O.

2. (a) Define natural frequency, normal mode of vibration, fundamental frequency, fundamental mode of vibration and resonance. (CO2)
- (b) A mass ' m ' is connected through 5 springs (k_1 and k_2 , k_3 , k_4 and k_5 in pairs in series). Determine the natural period : (CO2)

$$k_1 = 1000 \text{ N/cm}$$

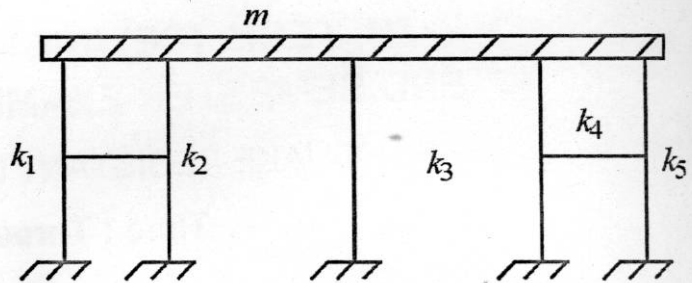
$$k_2 = 800 \text{ N/cm}$$

$$k_3 = 500 \text{ N/cm}$$

$$k_4 = 900 \text{ N/cm}$$

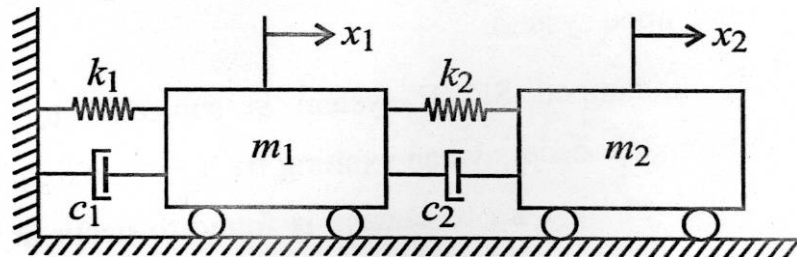
$$k_5 = 600 \text{ N/cm}$$

$$m = 2000 \text{ kg}$$



OR

- (c) Write down the equation of motion for the given multidegree of freedom system : (CO2)



3. (a) Explain the formation of seismic waves on the basis of elastic rebound theory. (CO3)
- (b) Differentiate between primary and secondary seismic waves with the help of diagrams. (CO3)

(3)

OR

- (c) Define the intensity of earthquake and explain how the intensity can be measured in Modified Mercalli Intensity Scale based on observations. Which factors influencing the intensity of earthquake at a particular place ? (CO3)
4. (a) Mention the role of response spectrum in designing earthquake resistant RC building. (CO4)
- (b) Differentiate between response spectrum and design spectrum. Mention the need for smoothened elastic design response spectrum. (CO4)

OR

- (c) Discuss the role of ductility in earthquake resistant design of buildings and how it can be improved in individual structural members as well as whole structure. (CO4)
5. (a) Mention the step-by-step procedure of determining the design base shear for a building using equivalent static lateral load method IS 1893 (part 1) : 2002. (CO5)
- (b) Discuss the mitigating effect of earthquake on structures. (CO5)

OR

- (c) Elaborate the purpose, principles and advantages of base isolation. (CO5)